

Responses to Cantonese A-not-A questions by Cantonese-English bilingual children

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Aims and objectives/purpose/research questions:

This study investigates mismatched responses to Cantonese A-not-A questions produced by Cantonese-English bilingual children. Mismatches are hypothesized to result from overgeneralisation, cross-linguistic influence and language dominance.

Design/methodology/approach:

Corpus data are drawn from eight Cantonese L1 children and eight Cantonese-English bilingual children from the Hong Kong Cantonese Child Language Corpus (Lee et al., 1996) and Hong Kong Bilingual Child Language Corpus (Yip & Matthews, 2007) respectively.

Data and analysis:

Children's production was sampled at three-month intervals from age 2;0 to 3;0. Responses to A-not-A questions were classified into 1) target response, 2) mismatch, or 3) others. Mixed effects logistic regression models were used to identify the effect of different variables.

Findings:

Two types of mismatches are observed: copular mismatch with *hai6/m4hai6* "be"/"be not" and existential mismatch with *jau5/mou5* "have"/"have not". Copular mismatches are attributed to cross-linguistic influence (CLI) and existential mismatches to overgeneralisation. No significant effect of language dominance is found.

Originality:

Although mismatched responses to A-not-A questions were mentioned in Chan (1995) and Lee et al. (2024), this is the first study to investigate the reasons behind this phenomenon.

Significance/implications:

This study provides evidence supporting Hulk & Müller's (2000) hypothesis from a novel domain. It demonstrates that cross-linguistic influence can occur from a weaker language to a dominant language in a vulnerable domain, and shows how syntactic and semantic models help predict non-target forms in bilingual children's production. The study contributes to the field of bilingual development by identifying a novel form of CLI, namely influence at the level of question-answer sequences.

Keywords

Response particle, A-not-A question, vulnerable domain, cross-linguistic influence, Cantonese

Introduction

Background

Cross-linguistic influence (CLI) refers to the direct or indirect effect of one language on the other (Serratrice, 2013). Hulk & Müller (2000) hypothesized two conditions for CLI to occur: the Interface condition and the Overlap condition. They argued that CLI takes place at the syntax-pragmatic interface and results from partial overlap in the surface levels of the two languages. Later studies (e.g., Nicoladis et al., 2021; Lee et al., 2024) show that CLI can occur at interfaces between the discourse-pragmatic interface and different linguistic modules (e.g., syntax, semantics and phonology). However, some occurrences of CLI arguably cannot be explained by the Overlap and Interface conditions (e.g., Pérez-Leroux et al., 2011).

Language dominance is one factor that influences the directionality of CLI (Yip & Matthews (2000). Cantonese-dominant children are found to produce prenominal relative clauses in English, manifesting CLI from Cantonese to the weaker language, English. Yip & Matthews (2007) found evidence of transfer from a non-dominant language (English) to a dominant language (Cantonese) only in certain vulnerable domains, that is, specific areas of grammar prone to CLI regardless of patterns of dominance, usually created by partial overlap where one language allows two possible structures but another allows only one of the two.

Previous studies on CLI mainly focus on morphology and syntax, for example, Cantonese-English bilingual children's acquisition of *wh*-questions (Yip & Matthews, 2000) and Spanish-English bilingual children's clitic placement (Pérez-Leroux et al., 2011). Some investigate CLI in other linguistic modules such as phonology, for example, Cantonese-English bilingual children's sentence-final prosody (Lee et al., 2024). Despite increasing attention to response particles and Cantonese-English bilinguals, there is no study on Cantonese-English bilingual children's responses to questions.

Previous studies on Chinese A-not-A questions and responses mostly focused on the syntactic derivations (e.g. Huang, 1991; Chen, 2022). Few studies have investigated children's acquisition of responses to A-not-A questions. In a study on children's interpretation of A-not-A questions, Chan (1995) elicited responses by asking *hai6-m4-hai6* questions. Incidentally, it was reported that 'a distinctive response type from Cantonese-speaking children was the indirect response: *yáu23* 有 "have" or *móuh* 無 "have not" + verb of the question' (Chan, 1995, p. 22). (1) is an example (shown in Jyutping romanization).

(1) Q: Keoi5 hai6-m4-hai6 siu3 aa3?

She be-not-be smile SFP

'Is it that she smiles?'

A: Jau5 siu3.

Have smile

'She has smiled.'

The target answer here is *hai6* “be” or *m4hai6* “be not”, but the child uses the existential verb *jau5* “have” as an auxiliary to refer to the past, such that the sentence “*jau5 siu3*” “she has smiled” is semantically relevant but not grammatically a target answer to the A-not-A question. This will be termed existential mismatch hereafter. Another type of non-target response is incidentally noted in Lee et al. (2024), where *hai6/m4hai6* “be”/“be not” is used by bilingual children as a response to A-not-A questions, even when *hai6* is not the A. (2) is an example.

(2) Q: Ngo5 ho2-m4-ho2ji3 zou6 Hermione aa3?

I can-not-can be Hermione SFP

“Can I be Hermione?”

A: M4 hai6 aa3.

Not be SFP

“No.”

The target answer is ho2ji3 “(you) can” or m4 ho2ji3 “(you) cannot” but the child uses m4 hai6 “be not” to express “no”. This response type will be termed copular mismatch. The present study investigates these two types of response, existential mismatch as in (1) and copular mismatch as in (2).

Responses to English yes/no questions

Yes/no questions, or polarity questions, are questions that expect polar answers (“yes” or “no”). An English yes/no question “Do you want tea?” is analysed as in (3) (Holmberg, 2016).

(3) [_{CP} Do [_{IP} you [_{±Pol}] want tea]]

The free variable [_{±Pol}] solicits an answer which assigns a polarity value to the IP. The invariant answer “yes”/“no” satisfies this requirement for most polarity questions. Krifka (2013) suggests that response particles like “yes”/“no” are anaphors which pick up a propositional discourse referent (i.e., the IP in the question in (3)) for assertion or negation. Holmberg (2016) argues that response particles to neutral polar questions cannot be anaphors because there are two referents (positive and negative) to pick up. He suggests an alternative ellipsis approach where an answer containing only a response particle is structured as (4).

(4) [_{FocP} yes/no [_{IP} ...X...]]

“Yes” or “no” occupies the Focus position and takes scope over the IP to assign a polarity value. In answering (3), “yes” takes sentential scope and assigns a positive polarity value to the IP, as in (5).

(5) [_{FocP} yes [_{IP} I want tea]]

(5) can then undergo IP-ellipsis to result in the surface form “yes”, or VP-ellipsis to give “yes, I do”.

Responses to Cantonese yes/no questions

Cantonese yes/no questions include particle questions, intonational questions, and A-not-A questions (Matthews & Yip, 2011). (6) and (7) exemplify the first two types, where the appropriate answer is *hai6* “be” or *m4hai6* “be not”.

(6) Q: Nei5 haa6 go3 lai5baai3 fong3 gaa3 aa4?

you next CL week take leave SFP

“You’re going on leave next week?”

A: Hai6

Be

“Yes.”

(7) Q: Nei5 zan1 hai6 heoi3?

you really go

“You’re really going?”

A: M4 hai6

Not be

“No.”

Particle questions (6) and intonational questions (7) turn a statement into a question by adding an interrogative particle or a rising intonation to the sentence-final position, structured as in (8).

(8) [CP [IP ...x...] interrogative SFP/intonation]

The Cantonese copular verb *hai6* “be” or its negative counterpart *m4hai6* “be not” can answer particle/intonational questions and declarative sentences. Previous studies on the syntactic structure of *hai6/m4hai6* “be”/“be not” (e.g., Cheng, 2022) and Mandarin copular verbs *shi/bushi* “be”/“be not” (e.g., Holmberg, 2016) as answers to yes/no questions are divided into anaphora-based and ellipsis-based approaches. Following Krifka (2013), Cheng (2022) argues that *shi* and *bushi* are anaphors which encode assertion and negation of the propositional discourse referent respectively. Cheng (2022) further explains that Holmberg’s (2016) objection to analysing “yes”/“no” as anaphor does not apply to *hai6* “be” because it is used to answer particle and intonational questions, which contain only one discourse referent each time for *hai6* “be” to pick up. If *hai6* “be” is an anaphor, it cannot answer a neutral polar question, e.g., A-not-A question, as it cannot pick up two discourse referents (+ and -) simultaneously. Following Krifka (2013), *hai6* “be” as an anaphor can be represented as (9).

(9) $\llbracket [\text{ActP } \textit{hai6}] \rrbracket = \text{ASSERT}(d)^1$

Alternatively, applying Holmberg’s (2016) ellipsis-based analysis to Cantonese *hai6/m4hai6* “be”/“be not”, the structure will be (10).

(10) $[\text{FocP } \textit{Hai6/m4hai6} [\text{IP } \dots \textit{x} \dots]]$

As Cheng (2022) points out, if *hai6/m4hai6* “be”/“be not” were structured as in (10), it should be able to answer neutral polar questions just like English “yes”/“no”. However, Cantonese *hai6/m4hai6* “be”/“be not” cannot answer neutral polar questions, i.e., A-not-A questions, unless it already appears in the question as A (Matthews & Yip, 2011: 370). We therefore adopt Cheng’s (2022) analysis treating *hai4/m4hai6* “be”/“be not” as an anaphor, serving to affirm/negate the propositional discourse referent.

Cantonese A-not-A questions are yes/no questions which involve reduplication of a verb or adjective, with a negation marker *m4* inserted. (11) is an example from Matthews & Yip (2011) and its target response.

- (11) Q: Gin6 saam1 leng3-m4-leng3 aa3?
 CL dress pretty-not-pretty SFP
 “Does the dress look good?”
 A: Leng3.
 pretty
 “Yes.”

According to Huang (1991), Holmberg (2016) and Chen (2022), the A-not-A construction is the overt realisation of polarity $[\pm\text{Pol}]$. The structure of the question in (11) is shown in (12).

(12) $[\text{CP } [\text{IP } \textit{CL dress} [\text{PolP } \textit{A-not-A pretty}]] \text{ SFP}]$

Such a neutral polar question cannot be answered with *hai6/m4hai6* “be”/“be not” in Cantonese. The answer to an A-not-A question typically consists of the verb alone, preceded by the negative marker *m4* for a negative answer (Matthews & Yip, 2011: 369). This is known as a verb echo answer (VEA). In answering (12), the VEA is *leng3* “pretty” or *m4leng3* “not pretty”.

Cross-linguistic influence

Both the Interface and Overlap conditions in Hulk & Müller’s (2000) hypothesis predict transfer in the production of responses to A-not-A questions by Cantonese-English bilingual children. First, response particles are situated at the syntax-pragmatic interface. Response particles occupy a syntactic position but have largely pragmatic functions (Holmberg, 2016, Mosegaard Hansen, 2020). Second, the Overlap condition applies: Cantonese yes/no questions allow both invariant

¹ d is the propositional discourse referent.

responses (i.e., *hai6* “be” and *m4hai6* “be not”) and variable responses (VEA) whilst English yes/no questions can only be answered with invariant responses (“yes”/“no”). There is thus partial overlap between the languages in responses to polar questions.

Following the syntactic discussion above, English “yes”/“no” and Cantonese *hai6*/*m4hai6* “be”/“be not” involve different structures: English “yes”/“no” occupies the focus position, taking scope over the IP which undergoes ellipsis (4), whilst Cantonese *hai6*/*m4hai6* “be”/“be not” is an anaphor (9) because it is limited to answering non-neutral polar questions (i.e., particle and intonational questions) and declarative sentences, which are either positive or negative in polarity. If *hai6*/*m4hai6* “be”/“be not” had the same structure as “yes”/“no”, it could answer all polar questions including neutral ones (i.e., A-not-A questions), which is ungrammatical in Cantonese. However, in the context of acquisition, when a bilingual child hears a Cantonese particle/intonational question or a declarative sentence answered by *hai6*/*m4hai6* “be”/“be not”, as in (6) or (7), this input is consistent with both the anaphora-based and the ellipsis-based analyses. Such input ambiguity in Cantonese favours the ellipsis-based analysis as it is the only structure consistent with the distribution of “yes”/“no” in English, leading to application of the English invariant response to Cantonese. Bilingual children’s misanalysis of the structure of *hai6*/*m4hai6* “be”/“be not” may lead them to use it in response to all Cantonese yes/no questions, including A-not-A questions as in (2) above.

However, answering A-not-A questions with invariant responses may not result entirely from CLI. Since particle and intonational questions in Cantonese allow responses with *hai6*/*m4hai6* “be”/“be not” (as in (6) and (7)), children might extend this response to A-not-A questions with *hai6*/*m4hai6* “be”/“be not” due to overgeneralisation. A third possibility is that overgeneralisation interacts with CLI, resulting in quantitative differences in the production of non-target *hai6*/*m4hai6* “be”/“be not” responses (i.e., copular mismatches) to A-not-A questions between bilingual and monolingual children.

Current Study

This study investigates the responses to A-not-A questions produced by Cantonese-English simultaneous bilingual children. By comparing the responses to A-not-A questions produced by Cantonese L1 and Cantonese-English bilingual children, we discuss how the invariant “yes”/“no” responses in English influence children’s responses to A-not-A questions. Our research questions are as follows:

- 1) If children produce mismatched responses to A-not-A questions, what types of mismatches are produced?
- 2) To what extent do bilingual and Cantonese L1 children produce target responses and mismatches to A-not-A questions?
- 3) Does language dominance profile of bilingual children affect their production rate of mismatches?

Hypotheses and predictions

Based on the literature review, we formulate the following hypotheses and predictions to address the above research questions.

1a) We predict that copular mismatch will be attested. Children will produce non-target *hai6/m4hai6* “be”/“be not” responses to answer A-not-A questions, whether resulting from overgeneralisation or CLI.

1b) We predict that children will produce non-target *jau5/mou5* “have”/“have not” (+ verb) responses to answer A-not-A questions as observed in Chan (1995).

2a) We hypothesize that overgeneralisation of *hai6/m4hai6* “be”/“be not” as response particle will occur, so that copular mismatches will be attested in the production of both Cantonese-English bilingual and Cantonese L1 children.

2b) We hypothesize that CLI will occur in bilingual children’s responses to A-not-A questions. We predict that Cantonese-English bilingual children will produce more copular mismatches and fewer target VEAs than Cantonese L1 children. If only CLI occurs but not overgeneralisation, we expect to see copular mismatches only in bilingual children’s production but not in Cantonese L1 children’s speech.

3) Given previous studies (e.g., Yip & Matthews (2000), Lee et al. (2024)) showing that language dominance influences the directionality and quantity of CLI, we hypothesize that the more Cantonese-dominant the child, the less CLI from English will occur, and the fewer copular mismatches to A-not-A questions will be produced.

Methods

Participants

Children’s production was sampled at three-month intervals from age 2;0 to 3;0. If there was more than one file at a timepoint, only the earliest file was examined.

We collected baseline data from eight Cantonese L1 children from the Hong Kong Cantonese Child Language Corpus (Cancorp) (Lee et al., 1996): Bohuen (wbh), Gakei (cgk), Bernard (mhz), Tsuntsun (ckt), Tinfan (ltf), Johnny (hhc), Jenny (lly), and Chunyat (ccc). Although often assumed to represent monolingual acquisition of Cantonese, these children are not strictly monolingual, since some have English exposure. Bernard was born in the UK before moving to Hong Kong at age 0;8 (Lee et al., 1996) and occasionally heard some English from his parents. One of Tinfan’s caregivers was a Filipino domestic helper, with whom Tinfan would communicate in “something English-like” (Lee et al., 1996). Jenny’s domestic helper would also speak English to her. Chunyat’s mother was a secondary school English teacher. Since it is common for middle-class families in Hong Kong to speak some English to children, especially when one parent is proficient in English, it is likely that Chunyat received English input. Similarly,

Johnny was looked after by a Thai domestic helper. It is common for domestic helpers in Hong Kong to speak English to children, or to code-mix using a considerable amount of English.

Therefore, during the time of data collection, five of the eight Cantonese L1 children have known or potential input from English. Thus, the Cancorp children are described as Cantonese L1 children, rather than monolingual, in this study. They serve as the baseline for comparison with bilingual children.

Table 1. Background information of the eight Cantonese L1 children

Child	Sex	Age of data used	No. of utterances	Cantonese MLU
Bernard (mhz)	M	2;00.03	375	2.128
		2;03.09	403	1.978
		2;06.04	304	2.674
Bohuen (wbh)	F	2;06.09	119	2.328
		2;09.26	344	2.495
Chunyat (ccc)	M	2;03.07	549	1.923
		2;06.08	681	2.167
		2;09.07	650	2.745
Gakei (cgk)	F	2;00.08	240	2.433
		2;03.04	451	2.9
Jenny (lly)	F	2;09.09	117	1.711
		3;00.11	594	3.227
Johnny (hhc)	M	2;06.10	648	1.872
		2;09.30	670	3.355
		3;00.08	618	2.331
Tinfan (ltf)	F	2;03.02	465	1.946
		2;06.01	611	2.957
		2;09.07	636	3.586
		3;00.20	634	2.517
Tsuntsun (ckt)	M	2;00.09	423	1.894
		2;03.03	691	2.339
		2;06.18	684	2.402

Cantonese-English bilingual children's data come from the Hong Kong Bilingual Child Language Corpus (Yip & Matthews, 2007). The production of eight Cantonese-English bilingual children was examined.² All eight children were exposed to both Cantonese and English from birth. At the time of data collection (1991-1994 for Cancorp, 1995-2007 for bilingual corpus), Putonghua, the standard form of Mandarin, was not taught in the children's kindergartens. Amongst the 16 children, only Timmy is reported to have received any Mandarin input: he had a Mandarin-speaking domestic helper from 0;3 to 0;6, but did not produce any Mandarin during the study period (Yip & Matthews, 2007). Given this limited exposure, a significant effect of Mandarin on the acquisition of responses to A-not-A questions in these children is unlikely; and any such effect would be qualitatively indistinguishable from the influence of Cantonese.

Language dominance of the bilingual children was determined by the child's MLU differential, direction of code-mixing, and language preference (Yip & Matthews, 2000). Since only Charlotte and Kasen after 2;10 are English-dominant, and only Darren and Kasen before 2;10 are balanced, these four children are grouped together as non-Cantonese-dominant children.

Both corpora recorded interactions between the children, the caregivers and investigators in naturalistic home settings. The investigators were research assistants and students hired for the development of the corpora in 1995-2007 (Yip & Matthews, 2007) and 1991-1994 (Lee et al., 1996). They were not instructed to elicit any kind of responses from children and were unaware of how their conversations with the children would be used, beyond the general aim of studying the children's language development. Three of the bilingual children were the children of authors of this study, one of whom participated in data elicitation for one of the children studied (Alicia). The Hong Kong Cantonese Child Language Corpus contains their interactions in Cantonese only, whilst The Hong Kong Bilingual Child Language Corpus contains both English and Cantonese contexts. Since this study aims to investigate their responses to Cantonese A-not-A questions, only recordings in Cantonese contexts are examined.

Table 2. Background information of the eight bilingual children

Child	Sex	Language dominance	Age of data used	No. of utterances	Cantonese MLU	English MLU
Alicia	F	Cantonese	2;00.13	68	2.794	2.714
			2;03.16	148	2.218	2.657
			2;06.02	172	3.152	2.139
			2;09.15	187	2.471	2.933
			3;00.10	234	3.41	2.435
Charlotte	F	Non-Cantonese	2;00.25	131	2.313	2.226
			2;03.17	152	2.5	2.309

² Kasen was originally balanced but turned English-dominant after 2;10 (Yip & Matthews, 2007). Kathryn is excluded as her Cantonese data begin at age three, which is beyond the age range of this study.

			2;06.16	161	2.472	2.663
			2;09.04	35	1.943	2.985
			3;00.03	123	3.024	4.072
Darren	M	Non-Cantonese	2;00.10	152	2.159	2.214
			2;03.08	148	2.453	2.587
			2;06.11	107	2.009	2.714
			2;09.03	242	2.755	3.064
			3;00.07	210	2.205	3.027
Janet	F	Cantonese	3;00.11	369	3.872	2.649
Kasen	M	Non-Cantonese	2;06.16	256	2.933	2.912
			2;09.03	369	3.309	2.723
			3;00.03	184	2.956	2.972
Llywelyn	M	Cantonese	2;00.12	149	1.439	1.559
			2;03.14	154	2.078	2.685
			2;06.20	300	3.267	3.439
			2;09.07	293	3.365	2.94
			3;00.27	292	2.969	3.49
Sophie	F	Cantonese	2;00.06	330	2.07	1.491
			2;03.01	237	2.848	1.954
			2;06.00	213	3.376	2.803
			2;09.06	361	3.438	2.974
			3;00.03	226	3.018	2.039
Timmy	M	Cantonese	2;03.17	325	3.434	3.362
			2;06.09	312	3.558	3.125
			2;09.08	217	3.82	3.739
			3;00.09	413	3.533	3.87

Procedure

All A-not-A questions addressed to the target children and their corresponding answers were extracted. Each response was judged as either 1) target response, 2) mismatch, or 3) others

(including ignored questions, English responses, and non-verbal responses). Target response means answering the A-not-A question with a VEA, that is, “A” for affirmation or “not A” for negation. Mismatched response means answering the A-not-A question with a word not used in the question. Such mismatches are judged ungrammatical by native speakers. Children’s responses which only repeated others’ speech were considered non-spontaneous and classified as “others”.

Results

Qualitative results

Mismatches in the children's production are categorized into 1) copular mismatch and 2) existential mismatch.

Copular mismatch

Copular mismatches involve using a Cantonese copula *hai6* ‘be’ or its negative counterpart *m4hai6* ‘be not’ to answer A-not-A questions in which the copula does not appear. They appeared in the production of both Cantonese L1 and bilingual children, as in (13) and (14) respectively.

- (13) INV³: Hou2sik6-mou2-sik6 gaa3
tasty-not-tasty SFP
‘‘Is it tasty?’’
CHI: Hai6 aa3.
be SFP
‘‘Yes.’’ (Jenny, 2;11.08)

In (13), the question uses the A-not-A construction *hou2sik6-mou2-sik6* ‘tasty-not-tasty’.⁴ The target VEA is *hou2sik6* ‘tasty’ to show affirmation (‘yes, it is (tasty)’). However, Jenny used the invariant response particle *hai6* ‘be’ to show affirmation.

- (14) LIN: Gam2 nei5 zung1-m4-zung1ji3 tai2 aa3?
 then you like-not-like watch SFP
 “So do you like to watch it?”
- CHI: Hai6 aa3.
 be SFP
 “Yes.” (Llywelyn, 2;06.20)

³ INV, LIN, WIN and MIC refer to the investigators, CHI the target child, FAT father, MOT mother, SIS sister, and HOU the housekeeper.

⁴ This construction is originally *hou2sik6-m4-hou2sik6*. The negative marker *m4* combines with the syllable *hou2* to form *mou2* in speech.

In (14), the question uses the A-not-A construction *zung1-m4-zung1ji3* “like-not-like”. The target VEA is *zung1ji3* “like” to show affirmation (“yes, I do (like to watch it)”). However, Llywelyn used the invariant response particle *hai6* “be” to show affirmation.

Existential mismatch

Existential mismatch involves using the existential verb *jau5* “have” or its negative counterpart *mou5* “have not” to answer A-not-A questions which do not use the existential verb. Children use these forms as responses to various A-not-A questions, not only those involving the status of an event as reported in Chan (1995). They appeared in the production of both Cantonese L1 and bilingual children.

(15) SIS: E4 Virginia hai2-m4-hai2 dou6?

eh Virginia at-not-at here

“Um, is Virginia here?”

CHI: Mou5.

have-not

“No.”

(Alicia, 2;06.02)

In (15), the question uses the A-not-A construction *hai2-m4-hai2* “at-not-at”. The target VEA is *m4-hai2* “not at” to show negation (“no, she is not (here)”). However, Alicia used the existential verb *mou5* “not have”. As the question in (15) asks about the existence of “Virginia”, this might encourage the child to use *mou5* to express her absence.

Existential mismatches are sometimes related to existential meaning in a less direct way, as in (16).

(16) LIN: Nei5 zeoi2seon4 hong4-m4-hong4 aa3?

your lips red-not-red SFP

“Are your lips red?”

CHI: Jau5 aa3.

have SFP

“Yes.”

(Llywelyn, 2;06.20)

In (17), Llywelyn answered *jau5* “have” instead of the target VEA *hong4* “red”. The mismatched response can be interpreted as “there is red on my lips”.

Other than producing existential mismatches in an existential-locative sense, they are sometimes used in a perfective sense, as in (17) and (18).

(17) INV: Hai6-mai6 jai5jai5 him1him1?

be-not-be naughty Johnny

“Are you naughty, Johnny?”

CHI: Mou5.

have-not

“No.”

(Johnny, 2;06.10)

(18) WIN: Hai6-mai6 faan1zo2 gung1 aa3?

be-not-be back ASP work SFP

“Is she back to work?”

CHI: Jau5 aa3.

have SFP

“Yes.”

(Llywelyn, 2;06.20)

In (17), Johnny answered *mou5* “have not” instead of the target VEA *m4hai6* “be not” to express the meaning “I have not been naughty”. In (18), Llywelyn answered *jau5* “have” instead of the target VEA *hai6* “be” to mean “she has gone to work”.

Quantitative results

As we found copular and existential mismatch in the production of both Cantonese L1 children and Cantonese-English bilingual children, we investigate whether their production of mismatches differ quantitatively.

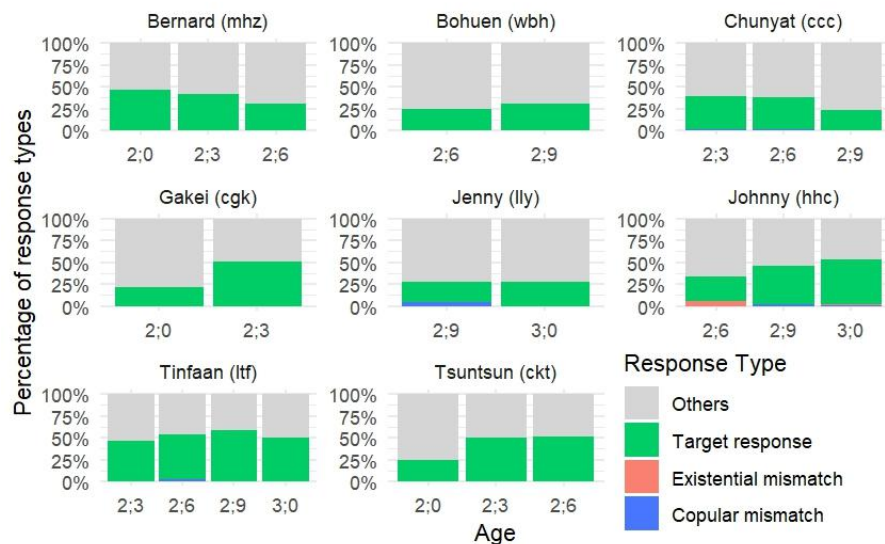


Figure 1. Percentage of responses by eight Cantonese L1 children

Mismatches are found in only four of the eight Cantonese L1 children, i.e., Chunyat, Jenny, Johnny, and Tinfan. Even for the four children who produced mismatches, their production of each type of mismatch occurs at relatively low frequencies (generally < 5% of responses).

Of the four children who produced no mismatches, Bohuen, Gakei, and Tsuntsun are not reported to receive any English input. All four children who produced mismatches produced copular mismatches. These four children were reported or suspected to have some exposure to English (see *Participants*).

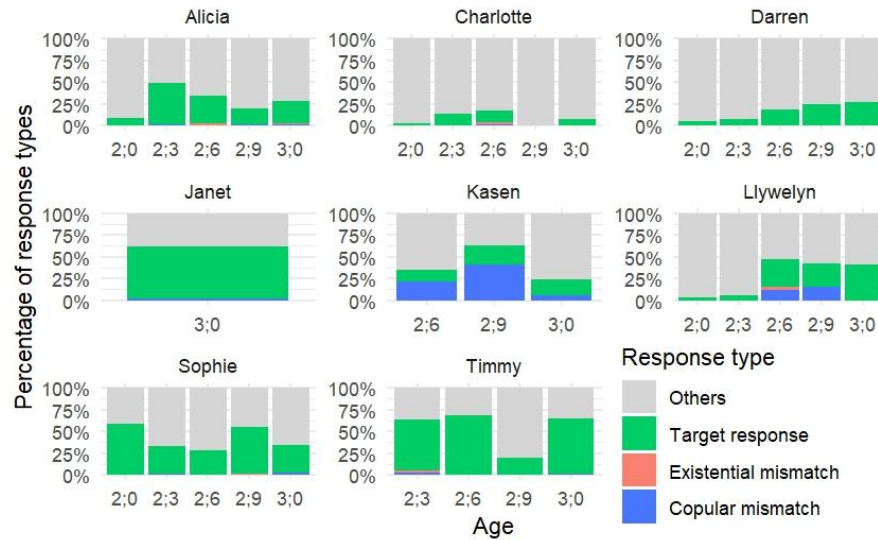


Figure 2. Percentage of responses by eight bilingual children

All bilingual children except Darren produced mismatches. All bilingual children who produced mismatches produced copular mismatches. Copular mismatch is unattested only in Darren's production in the Bilingual Corpus. Five children (Alicia, Charlotte, Llywelyn, Sophie, and Timmy) produced existential mismatches. Similar to Cantonese L1 children, existential mismatch is attested at low frequencies (< 5%) when compared to copular mismatch, which constitutes the majority of mismatches.

Copular mismatch is attested most frequently in the non-Cantonese dominant bilingual child, Kasen, who produced it in nearly 40% of all his responses (21 instances) at 2;9. Other non-Cantonese dominant children produced copular mismatches at lower rates. Only one instance of copular mismatch appears in Charlotte's production, while Darren did not produce any mismatches. The exceptionally low frequency of mismatch found in the production of these two children may be attributed to the overall paucity of their responses to A-not-A questions: Charlotte responded to only 8.3%, and Darren to 13.6% of A-not-A questions (Figure 2).

The Cantonese-dominant child Llywelyn is second to Kasen in the quantity of copular mismatches. 12% and 15% of his responses are copular mismatches at 2;6 and 2;9 respectively. In the remaining bilingual children, copular mismatch is attested at a low rate (< 5%) similar to the four Cantonese L1 children.

To test for quantitative differences in the production of mismatches between Cantonese L1 and bilingual children, we conducted four separate mixed-effects logistic regression analyses using lme4 package (Bates et al., 2015) in R (R Core Team, 2024). There were four dependent variables, with one in each model, including 1) target responses, 2) mismatches (copular + existential mismatches, 3) copular mismatches, and 4) existential mismatches). We included bilingualism (binary: bilingual vs. Cantonese L1) as the predictor, and participants as random intercepts.

Results show that the percentage of target responses in Cantonese L1 children's production is higher than in bilingual children (97.69% vs 87.93%). The percentage of mismatches in bilingual children's production is about five times of that of Cantonese L1 children (11.76% vs 2.31%). Cantonese L1 children produced significantly more target responses ($\beta = -2.17$, $SE = .85$, $z = -2.53$, $p = .01$) and fewer mismatches ($\beta = 2.12$, $SE = .86$, $z = 2.47$, $p = .01$) than bilingual children. Bilingual children produced copular mismatches at a rate of 10.41%, significantly higher than in Cantonese L1 children (1.10%, $\beta = 2.23$, $SE = .84$, $z = 2.64$, $p = .008$). By contrast, no significant difference is found in the production of existential mismatches between bilingual and Cantonese L1 children (1.20% vs 1.21%).

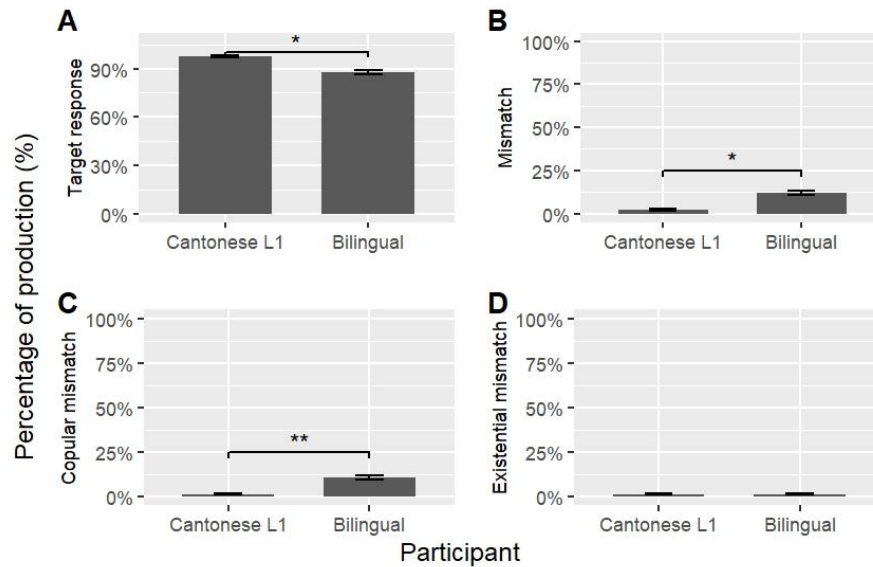


Figure 3. Production of (A) target response, (B) mismatch, (C) copular mismatch, (D) existential mismatch by Cantonese L1 and bilingual children

To investigate the effect of language dominance on the production of mismatches in bilingual children, we performed another four mixed-effects logistic regressions, using the same four dependent variables, one in each model: 1) target responses, 2) mismatches (copular + existential mismatches, 3) copular mismatches, and 4) existential mismatches). We included language dominance (binary: Cantonese-dominant vs. non-Cantonese-dominant) as the predictor, and participants as random intercepts.

There was no significant difference in the production of target responses ($\beta = -.88$, $SE = 1.06$, $z = -.84$, $p = .40$) or mismatches ($\beta = .95$, $SE = 1.06$, $z = .90$, $p = .37$) between Cantonese-dominant

and non-Cantonese-dominant children. Within the production of mismatches, no significant difference is found in the production of copular mismatches between Cantonese-dominant and non-Cantonese-dominant children ($\beta = .99$, $SE = 1.16$, $z = .85$, $p = .40$), or in the production of existential mismatches ($\beta = -.47$, $SE = 1.10$, $z = -.42$, $p = .67$). Although Figure 4 shows that non-Cantonese-dominant children produce fewer target responses (66.39%) and more copular mismatches (32.79%) than Cantonese-dominant children (target responses 92.79%; copular mismatches 5.36%), these results are disproportionally affected by the non-Cantonese-dominant child Kasen, who showed a peak rate of 40% copular mismatch at 2;9. As shown in Figure 2, the other two non-Cantonese-dominant children, Charlotte and Darren, produced copular mismatches at a low rate similar to Cantonese-dominant children. Therefore, a general effect of language dominance on the children's production of mismatches is not observed.

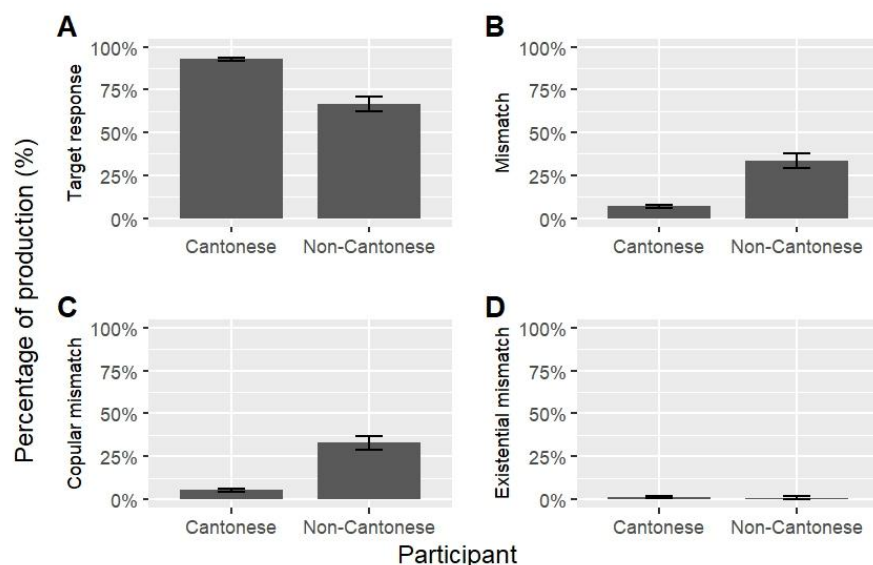


Figure 4. Production of (A) target response, (B) mismatch, (C) copular mismatch, (D) existential mismatch by Cantonese and non-Cantonese-dominant children

Discussion

The results support hypotheses 1a and 1b, which predict the occurrence of copular and existential mismatch. Although children mostly produced target VEAs and produced mismatches less than 15% of the time, the distribution of the two types of mismatch suggests that existential mismatch arises from overgeneralisation whilst copular mismatch stems from CLI, as discussed below.

Copular Mismatch

The results refute hypothesis 2a and support hypothesis 2b: there is evidence for CLI in children's responses to Cantonese yes/no questions, but not for overgeneralisation of *hai6/m4hai6* “be”/“be not” as a response particle. Copular mismatches are attested in all but one of the bilingual children but only half of the Cantonese L1 children, providing evidence for CLI in responses to Cantonese

yes/no questions. The lower rate of target responses and higher rate of copular mismatches in bilingual children’s production match our predictions. Copular mismatches are also found in the production of Cantonese L1 children known or presumed to have some English exposure, but not in those without known English exposure (i.e., Bohuen, Gakei, and Tsuntsun). Therefore, CLI is the most parsimonious explanation for copular mismatch, even in those four Cantonese L1 children who produce it, as they receive at least some English input.

In adult English, “yes”/“no” has an ellipsis-based structure as in (4) above. It answers both neutral and non-neutral polar questions, taking scope over the IP in the question and assigning a polarity value to it. Contrastively, in adult Cantonese, *hai6/m4hai6* “be”/“be not” bears an anaphora-based structure as in (9). As an anaphor, it can only pick up one propositional discourse referent, so it cannot be used to answer neutral polar questions such as A-not-A questions, which contain two. It answers only non-neutral polar questions (i.e., particle and intonational questions) and declarative sentences in Cantonese. Children receive input where *hai6/m4hai6* “be”/“be not” is used to answer particle/intonational questions and declarative sentences, such as (19).

- (19) MIC: Nei5 zoek3 zyu6 tiu4 kwan4kwan2 aa4? Gei2 leng3 bo3.
 you wear PRT CL dress SFP quite pretty SFP
 “You wearing a dress? It’s quite pretty.”
 MOT: Hai6 aa3.
 be SFP
 “Yes.” (Alicia, 1;03.10)

In cases like (19), *hai6/m4hai6* “be”/“be not” is compatible with both the ellipsis-based structure as in (4) and the anaphora-based structure as in (9). Since “yes”/“no” bears an ellipsis-based structure in English, bilingual children might also apply the ellipsis-based analysis to *hai6/m4hai6* “be”/“be not”, as in (10), allowing it to answer A-not-A questions (Figure 5).

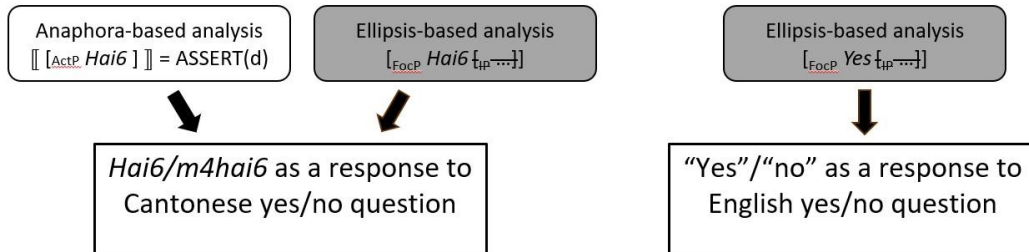


Figure 5. Input ambiguity in Cantonese under English influence

Moreover, *hai6/m4hai6* “be”/“be not” can be the target VEA to A-not-A questions using the copular verb as A, such as the question in (17). Children may not differentiate between *hai6/m4hai6* “be”/“be not” as an invariant response and as a VEA in such cases. As a result, the ambiguous *hai6/m4hai6* “be”/“be not” response can accommodate two different analyses (Figure 5). The structure of “yes”/“no” responses in English provides additional motivation for adopting the ellipsis-based analysis for *hai6/m4hai6* “be”/“be not” in Cantonese.

This argument assumes that bilingual children produce “yes”/“no” in English such that they can transfer its ellipsis-based structure to *hai6/m4hai6* “be”/“be not”. It is found that all bilingual children produce “yes”/“no” as early as their first sampled file, as in (20).

- (20) WIN: Is it nice?
CHI: Yes. (Sophie, 1;11.08)

In Figure 5, children correctly apply the ellipsis-based analysis to English “yes”/“no”, instead of having competing analyses as in the case of *hai6/m4hai6* “be”/“be not”, because the English input is unambiguous. Children receive plentiful input where “yes”/“no” answers neutral polar questions like (21), serving as evidence against the anaphora-based analysis.

- (21) LIN: Can rabbit jump?
HOU: Yes. (Llywelyn, 2;00.12)
- (22) CHI: Wash the this one.
FAT: Yes, I wash that one. (Alicia, 2;03.02)

Cases like (21) demonstrate to children that “yes” can serve as an answer to neutral polar questions, so it does not bear an anaphora-based structure. (22) even shows an optional IP following “yes”, providing direct positive evidence for the ellipsis-based analysis of “yes”/“no” as in (9) above. Therefore, it may be easier for children to acquire the ellipsis-based structure of “yes”/“no” than the anaphora-based structure applicable to Cantonese.

Existential Mismatch

As observed in *Results*, children use the existential verb *jau5/mou5* “have”/“have not” to express location and existence, as well as past events. Their use of existential verbs matches the questions semantically: the basic use of *jau5/mou5* “have”/“have not” is to express existential-locative meanings. All children produce existential-locative sentences with *jau5/mou5* “have”/“have not”, as in (23).

- (23) **Jau5** bo1.
have ball
“There is a ball.” (Alicia, 2;00.26)

This explains why the children used the existential verb *jau5/mou5* “have”/“have not” to express existential-locative meanings as in (15) and (16). Although their responses do not follow the grammatical form of VEA, they match the questions semantically. For example, we can interpret the child’s utterance in (16) as (24):

- (24) Ngo5 zeoi2seon4 **jau5** hong4 sik1.
my lips **have** red colour
“There is red on my lips.”

Jau5/mou5 “have”/“have not” in Cantonese can also indicate a present state which results from an action undertaken in the past (Matthews & Yip, 2011: 237). This use is found in all children’s production, especially when responding to A-not-A questions that use existential *jau5/mou5* “have”/“have not” in this way, as in (25).

- (25) INV: *Jau5-mou5 bei2 maa1mi4 daa2 aa3?*
 have-not-have by mommy beat SFP
 “Have you got beaten by mommy?”
 CHI: **Mou5.**
have-not
 “No.” (Johnny, 2;06.10)

To answer questions asking about the state/result of an event/action, the children occasionally answer with *jau5/mou5* “have”/“have not” although it is not the target VEA: in (17), the child used *mou5* to express that no action of being naughty had occurred. We can interpret (17) as (26).

- (26) Ngo5 **mou5** jai5.
 I **have-not** naughty
 “I have not been naughty.”

Similarly, in (18), the child used *jau5* “have” to express that his mother had gone to work. These answers would be grammatically correct if the questions had used the A-not-A construction *jau5-mou5* “have-not-have”, as in (25). The occurrence of existential mismatch may be attributed to the frequent use of *jau5/mou5* “have”/“have not” to answer Cantonese yes/no questions in which these morphemes appear (as in (25)). Therefore, it is the result of overgeneralisation of the existential verb as a response. Since English influence is not involved, there are no differences in the production of existential mismatches between Cantonese L1 children and bilingual children, or within bilingual children.

Language Dominance

Since no significant effect of language dominance on the production of target responses or mismatches is found, the findings do not directly support hypothesis 3 that there will be less CLI from English in more Cantonese-dominant individuals. However, we do observe a trend supporting it: non-Cantonese-dominant children produced 33% copular mismatches while Cantonese-dominant children produced 5%. The statistically non-significant result reflects the small sample size and individual variation.

Some confounding factors affect the production of target VEAs and the occurrence of CLI. Within the non-Cantonese-dominant group, there are English-dominant (Charlotte and Kasen after 2;10) and relatively balanced children (Darren and Kasen before 2;10). Despite Charlotte’s dominance in English, we only observed one instance of copular mismatch in her production. This can be attributed to her low frequency of responding to A-not-A questions in Cantonese (8.3%) and her habitual code-switching: she often ignored questions or answered in English. As aforementioned, Darren did not produce any tokens of mismatch. He responded to A-not-A questions at a low

frequency (13.6%) compared to other bilingual children (generally > 30%). The absence of mismatches can be attributed to the paucity of responses in his production. As a result, Kasen (in the period 2;0–2;10) is the only typical balanced child in this study, so we cannot generalize his production pattern to all balanced children. Therefore, we cannot confirm the effect of language dominance on the production of copular mismatches.

Table 3. Summary of research questions, predicted outcomes and results.

Research questions	Predictions	Results
1. What types of mismatches are produced?	Copular mismatch will occur. Existential mismatch will occur.	Copular mismatch occurs. Existential mismatch occurs.
2. To what extent do bilingual and Cantonese L1 children produce target responses and mismatches to A-not-A questions?	Overgeneralisation will occur, so that copular mismatches will be produced by both bilingual and Cantonese L1 children. CLI will occur in bilingual children's production so bilingual children will produce more copular mismatches and fewer target VEAs than Cantonese L1 children.	Bilingual children produced more copular mismatches and fewer target VEAs than Cantonese L1 children. Copular mismatch is only attested in the production of children with English exposure but not Cantonese L1 children without reported English exposure. Therefore, only CLI occurs, but not overgeneralisation.
3. Does language dominance play a role in bilingual children's production of mismatches?	The more Cantonese-dominant the child, the less CLI from English will occur, and the fewer copular mismatches will be produced.	No significant effect of language dominance on the production of copular mismatch is found.

As reviewed in *Introduction*, existential and copular mismatches were incidentally found in Chan (1995) and Lee et al. (2024) respectively. Cantonese-speaking children were found to respond to *hai6-m4-hai6* questions with *jau5/mou5* “have”/“have not” (+ verb) in Chan (1995), mirroring the existential mismatches found in our study, e.g., (17), (18), and (25), where *jau5/mou5* “have”/“have not” were used to answer A-not-A questions in which the existential verb is not the A. As argued in *Existential Mismatch*, these are cases of semantically-driven overgeneralisation of the verb *jau5/mou5* “have”/“have not”. We can now explain why Chan (1995) observed cases of existential mismatch, but not copular mismatch. Since she elicited responses from children by

posing *hai6-m4-hai6* questions, *hai6/m4hai6* “be”/“be not” was the target VEA. Therefore, no copular mismatch could be found. Additionally, we found that copular mismatch arises from CLI instead of overgeneralisation, explaining why copular mismatch could be observed in Lee et al.’s (2024) bilingual data but not in Chan’s (1995) monolingual data.

This raises another question: why does overgeneralisation occur with the existential *jau5/mou5* “have”/“have not” but not with the copula *hai6/m4hai6* “be”/“be not”? The overgeneralisation of existential verb is semantically-motivated: A-not-A questions related to event state and existential-locative meaning motivate children to respond with *jau5/mou5* “have”/“have not”. By contrast, overgeneralisation of the copula would entail extending the role of *hai6/m4hai6* “be”/“be not” from a response to other yes-no questions to a response to A-not-A questions. Such overgeneralisation presupposes that Cantonese L1 children cannot yet distinguish A-not-A questions from other yes-no questions and have not acquired an anaphora-based structure for *hai6/m4hai6*. The results, however, suggest that they could. Whether this is the case can be tested experimentally in future studies.

Conclusion

This is the first study to investigate a phenomenon noticed incidentally in Chan (1995) and Lee et al. (2024): mismatched responses to A-not-A questions in child Cantonese. We found two types of mismatches in the children’s production: copular and existential mismatch. Copular mismatches occur consistently in bilingual children, and in Cantonese L1 children with known or suspected exposure to English. These results suggest that copular mismatch is attributable to CLI rather than overgeneralisation. Despite the low frequency of mismatches, their distribution patterns reveal that copular mismatch arises from CLI and occurs exclusively among children with English exposure. By contrast, existential mismatch is attributed to overgeneralisation of *jau5/mou5* “have”/“have not” in response to Cantonese yes/no questions.

This study contributes to the field of bilingual development by identifying a novel form of CLI, namely influence at the level of question-answer sequences. This instance of CLI takes place at the interface of syntax and discourse, and can be attributed to overlap between the response types of the two languages, thus meeting the Interface and Overlap conditions proposed by Hulk & Müller (2000).

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